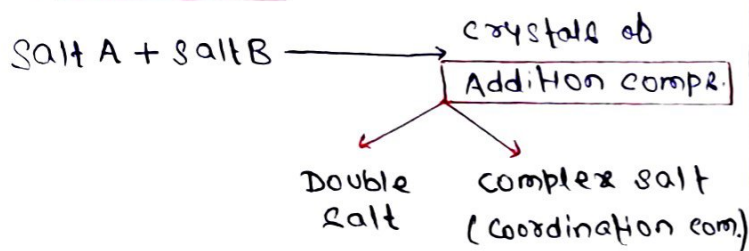


COORDINATION COMPOUND

Introduction:-



Double salt

- Completely ionise in aq. medium.
- Loose their identity in aq. solⁿ.
- Each ion gives positive test
- Individual ions do not loose their own identity.

Examples

- ① Carnallite $[KCl \cdot MgCl_2 \cdot 6H_2O]$
- ② Mohar's salt $[FeSO_4(NH_4)_2SO_4 \cdot 6H_2O]$
- ③ Potash alum $[K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O]$

Complex salt

- Do not ionise completely.
- Retain their identity in aq. medium
- Each ion do not give positive test
- Ions loose their own identity.

Examples:- $K_4[Fe(CN)_6]$, $[Pt(NH_3)_2Cl_2]$ etc.

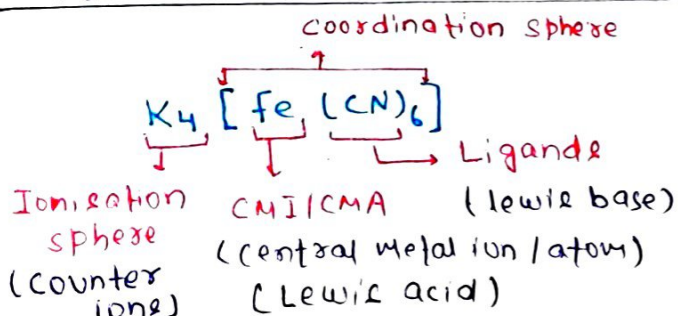
$\rightarrow$ Complex salt $\rightarrow$ σ coordinate bonds present b/w ligand & (e^- pairs donor) & Metal ion & (e^- pairs acceptor).

$\Rightarrow$ Why d-block element (Metal) form more no. of complexes?

Reason:- ① Availability of vacant orbital

② Small size & High ϕ nature

$\phi \uparrow$, Covalent nature of Bond $\uparrow$



Definitions

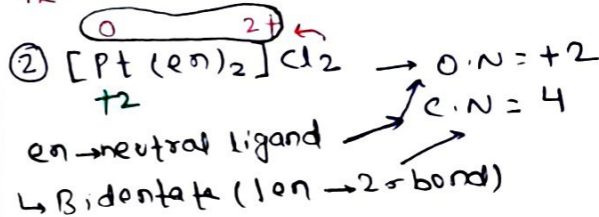
① Coordination number:- No. of ligand donor atoms to which metal is directly bonded. or

$\rightarrow$ Coordination no. is equal to σ -bonds present b/w ligand & CMI.

According to NCERT, π -bond b/w CMI & ligands will not be included in CN.

② Oxidation no. of central atom:-

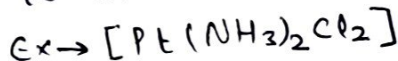
$\rightarrow$ Andaz & Bahar ke charge ka Hi saab



③ Homoleptic complex:- CMI is bound to only one kind of ligands.



④ Heteroleptic complex:- CMI is bound to more than one kind of ligands.



⑤ Coordination polyhedron:- Dabbe ki geometry (decided by CN of complex)

(C.N)

2 $\rightarrow$ linear

3 $\rightarrow$ T. Planar

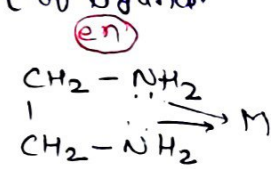
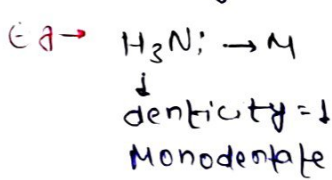
4 $\rightarrow$ Sq. Planar

* 4 $\rightarrow$ Tetrahedral

5 $\rightarrow$ T. B. P

* 6 $\rightarrow$ S. B. P or Octahedral

Denticity:- no. of σ -coordinate bond formed by one unit of ligand.



2 σ -coordinate bond by one unit

d = 2 Bidentate.

(Jugad) If any ligand, its total no. of atoms are 3 or less than 3, then denticity is always Monodentate.

Ligands :- Molecules/ions capable of donating e^- pairs to CMT.

Classification of Ligands

on the basis of charge	on the basis of denticity	Based on e^- donation/accept
Anionic	Monodentate ($d=1$)	Classical
Neutral	Polydentate ($d>1$)	Non-classical
Cationic	Ambidentate	
	Flexidentate	

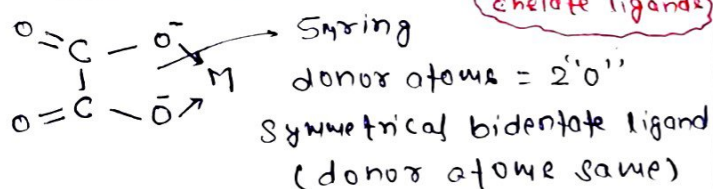
① On the Basis of charge

A) Anionic (Subfix = -)

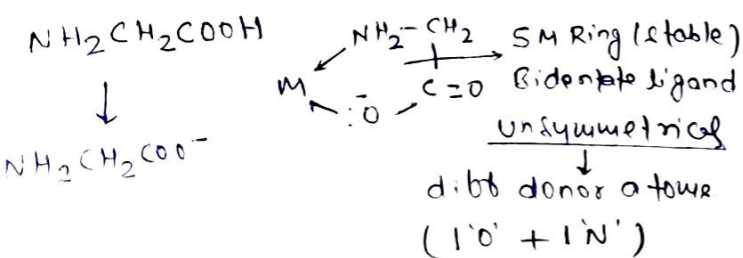
- | | |
|-----------------------------|--------------------------------|
| (1) SO_4^{2-} - Sulphato | (12) O_2^- - Superoxido |
| (2) CO_3^{2-} - Carbonato | (13) O_2^{2-} - Peroxido |
| (3) Cl^- - Chlorido | (14) CN^- - Cyanido |
| (4) Br^- - Bromido | (15) NC^- - Isocyanido |
| (5) OH^- - Hydroxido | (16) NO_2^- - Nitrito - N |
| (6) F^- - Fluorido | (17) ONO^- - Nitrito - O |
| (7) NH_2^- - Amido | (18) OCN^- - Cyanato - O |
| (8) NH_2^- - Imido | (19) NCO^- - cyanato - N |
| (9) N_3^- - Azido | (20) SCN^- - Thiocyanato - S |
| (10) N_3^- - Nitrido | (21) NCS^- - Thiocyanato - N |
| (11) O^{2-} - Oxido | |

1 to 21 → Monodentate
14 to 21 → Ambidentate

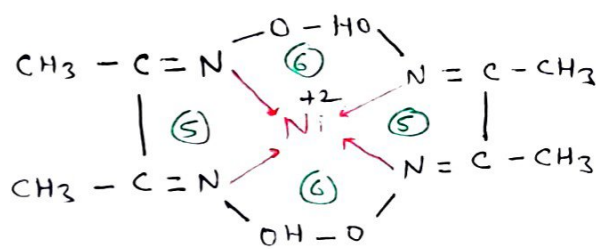
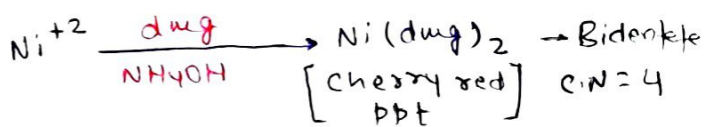
(22) $C_2O_4^{2-}$ (Oxalato or ox)



(23) Gly^- = Glycinate



(24) dmg^- (dimethyl glyoximate)



(Nickel dimethyl glyoxime complex)

Imp. points about $[Ni(dmg)_2]$ complex

- ① Formed during test of Ni^{2+} ions.
 - ② Rosy Red / cherry Red ppt.
 - ③ sq planar geometry ($c.n=4$), Ni is $d sp^2$ hybridised.
 - ④ Optically Inactive complex (POSV)
 - ⑤ Donor atom → N
 - ⑥ Total Rings → 4 (Two 6M + Two 5M)
 - ⑦ Symmetrical H-bonding Present.
- (25) $(EDTA)^{4-}$ (Ethylenediamine tetraacetate)

- Hexadentate ($d=6$)
 → 2 'N' & 4 'O' donor atoms.
 → Total rings → 5
 Each is 5-membered
 → Non-coplanar rings are present
 ∴ optically Active

B) Neutral ligands

- | | |
|-------------|---|
| Monodentate | (1) H_2O → Aqua |
| | (2) NH_3 → Ammine |
| | (3) CO → Carbonyl |
| | (4) NO → Nitrosyl |
| | (5) PH_3 → Phosphine |
| | (6) PPh_3 → Triphenyl phosphine |
| | (7) Py → Pyridine |
| Polydentate | (8) en → ethylenediamine ($d=2$) |
| | (9) pn → propylene diamine ($d=2$) |
| | (10) tn → trimethylene diamine ($d=2$) |
| | (11) dien → diethylene triamine ($d=3$) |
| | (12) trien → triethylene tetraamine ($d=4$) |

donor atoms (N)

C cationic ligands

- (1) $\text{NO}^+ \rightarrow$ Nitrosonium
- (2) $\text{NH}_2\text{-NH}_3^+ \rightarrow$ Hydrazenium

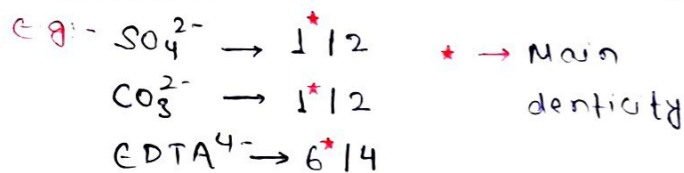
② on the Basis of denticity

- | | Anionic | Neutral |
|---------------------------|------------|-----------|
| (1) Monodentate ($d=1$) | (1 to 21) | (1 to 7) |
| (2) Polydentate ($d>1$) | (22 to 25) | (8 to 12) |

(14 to 21)

(3) Ambidentate:- Those ligands which have more than 1 donor atoms but all are not simultaneously functional.

(4) Flexidentate:- variable denticity



(3) Classification based on electron donation / Acceptance:-

① Classical $\rightarrow$ e^- pair donor (LP donor)

② Non-classical $\rightarrow$ e^- pair donor as well accept e^- density from Metal.

• π Acid ligands $\rightarrow$ form π bond by accepting e^- density from CMA.

• Imp. Exm:- $\text{CH}_2=\text{CH}_2$, $[\text{Co}]$, CN^- , C_6H_6 etc.

IUPAC NOMENCLATURE of Coordination comp.

(1) Name the cation first.

(2) For cationic $[]^+$ or neutral $[]^0$ complexes.
 no. of ligands + Name of ligands
 (Bi, tri, tetra, penta) (Alphabetical order)
 Hexa etc.

+ Name of Metal + O.N of Metal in Roman enclosed in Bracket $[]$

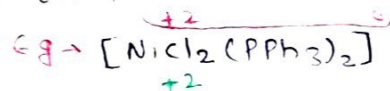
(3) For anionic complex $[]^-$ add 'ate' suffix with Metal.



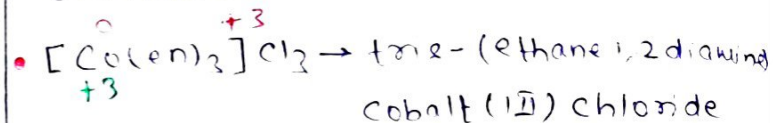
Rest all rules are same as above.

Eg:- Fe $\rightarrow$ Ferrate
 Zn $\rightarrow$ Zincate

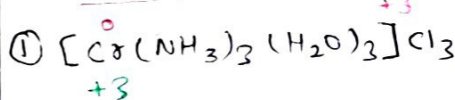
$\rightarrow$ When the names of the ligands include a numerical prefix then the terms -bis, -tris, -tetrakis are used.



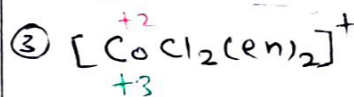
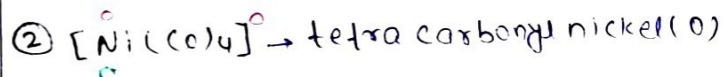
Dichlorobis (triphenyl phosphine) Nickel (II)



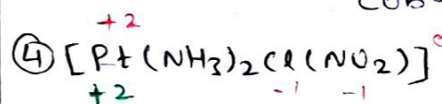
Q $\Rightarrow$ IUPAC Practice



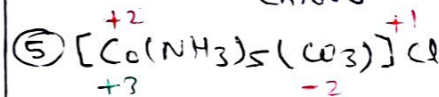
$\rightarrow$ Triammine triaquachromium (III) chloride



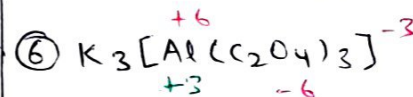
$\rightarrow$ dichlorido bis (ethane 1,2 diamine) cobalt (II) ion



$\rightarrow$ diammine chlorido nitrito-N Platinum (II)



$\rightarrow$ Pentaammine carbonate cobalt (II) chloride

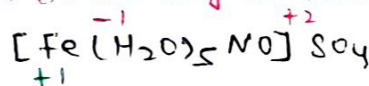


$\rightarrow$ Potassium trioxalatoaluminate (III)

Special cases

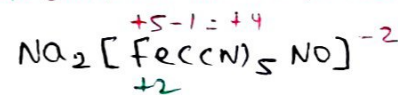
$\rightarrow$ When Fe & NO are together in complex then NO will be Not $\rightarrow$ Nitrosonium

$\rightarrow$ Brown ring complex $\rightarrow$ test of NO_3^-



$\rightarrow$ Penta aqua nitrosonium iron (I) sulphate

$\rightarrow$ Sodium Nitroprusside $\rightarrow$ test of S^{2-} ion



$\rightarrow$ Sodium Penta cyanido nitrosonium ferrate (II)

$\text{S}^{2-} \xrightarrow{\text{SNP}} \text{purple colour}$

Bonding in complexes

- ① Werner's Theory
- ② VBT
- ③ CFT (Crystal field theory)

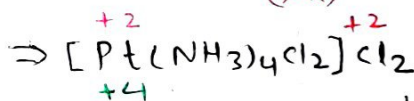
① Werner's theory:- CMI/CMA have two types of valencies.

1° valency

- Equal to O.N of CMI
- Satisfied by anions
- Ionisable
- Non-directional
- Represented by ... line
- Geometry prediction (xx)

2° valency

- equal to C.N of CMI
- Satisfied by neutral & anions.
- Non-ionisable
- Directional (Covalent bonded)
- by — solid lines
- Geometry prediction (✓✓)



→ O.N / 1° valency = 4

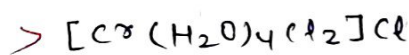
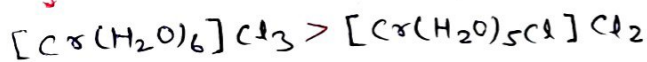
Primary valency → 4

But ionisable → 2

→ On rxn with excess of $AgNO_3$, No. of $AgCl$ moles produce → 2 $AgCl$ ↓ PPT

→ Jitna Cl^- bahar uttha $AgCl$ ka ppt

→ Molar conductivity & no. of ions order



② VBT (Valence bond theory)

→ Geometry, hybridisation etc in a complex can be predicted from coordination no. of metal ion.

→ Vacant orbitals of CMI undergo into hybridisation & use them to form coordinate bonds with ligands.

C.N Geometry Hybridisation

- 2 Linear → SP
- 3 T. Planar → SP^2
- 4 Tetrahedral → SP^3
- 4 sq. planar → $dSP^2 \rightarrow dx^2-y^2$
- 5 T. Bipyramidal → SP^3d / $dSP^3 \rightarrow dz^2$
- 6 sq. / octahedral → SP^3d^2 / $d^2SP^3 \rightarrow dz^2, dx^2-y^2$

* Donor strength

Strong ligand (C > N)

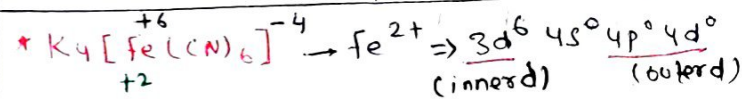
- Can pair d'e
- diamagnetic
- Inner orbital complex
- low spin

Generally

Weak ligand (O > X)

- can not pair d'e
- paramagnetic
- outer orbital complex
- High spin

Generally

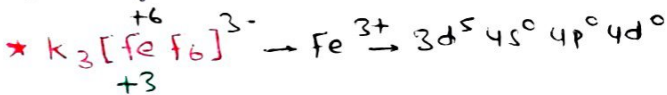


$CN^- \rightarrow$ strong ligand

Pairing ✓

C.N = 6

$d^2 sp^3$, inner, $n=0$, diamagnetic, low spin diamagnetic.

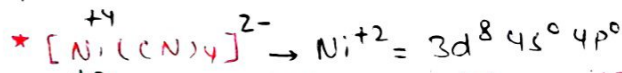


$F^- \rightarrow$ weak ligand

Pairing ✗

C.N = 6

$sp^3 d^2$, octahedral, outer, $n=5$ ($u=5, \dots$), paramagnetic, High spin.

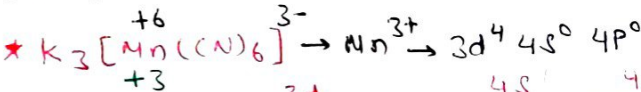


$CN^- \rightarrow$ S.L

Pairing ✓

C.N = 4

dsp^2 , sq planar, inner, $n=0$, diamagnetic



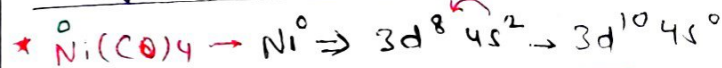
$CN^- \rightarrow$ S.L

Pairing ✓

C.N = 6

octahedral, low spin, inner $n=2$ ($u=2, \dots$), Paramagnetic

* Pairing utni hi karwani hai jitni required h.



transfer of ~~4~~ 4s e⁻ into 3d

C.N = 4

SP^3 , tetra, outer, $n=0$, diamagnetic

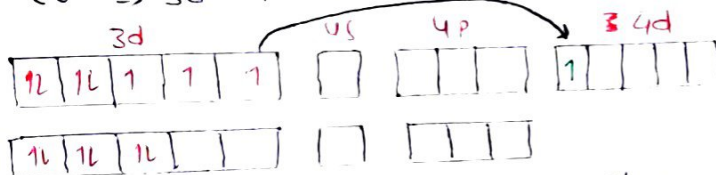
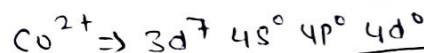
Jugad

S.L/W.L

$d^1/d^2/d^3$ 6 ligands d^2sp^3 always inner d
1... 2... 3... (C.N=6) Paramagnetic

$d^8/d^9/d^{10}$ 6 ligands sp^3d^2 always outer
Para para dia (C.N=6)

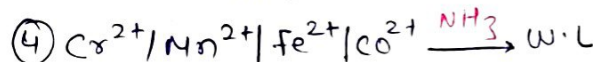
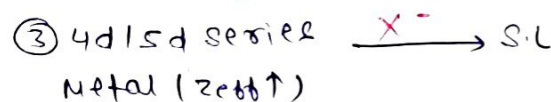
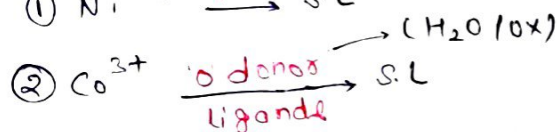
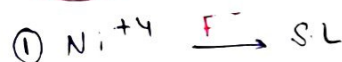
Transfer Rule - If required 1 electron can be excited to high energy orbital, but in presence of strong ligand only.



$\rightarrow d^2sp^3$, octa, $n=1$, Para, inner d, low spin

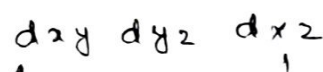
Strength Rule - (Special cases)

$(M+4) \rightarrow$ All ligands work as S.L



③ CFT (Crystal Field Theory)

- Based on electrostatic model
- Consider metal cation as a point '+' charge & ligand as point '-' charge.
- When ligand approaches metal ion, energy of d-orbital increases (due to electronic repulsion).
- Increase in energy of d is unsymmetrical which causes splitting in d-orbitals.
- Nature of splitting is depend on type of attack.

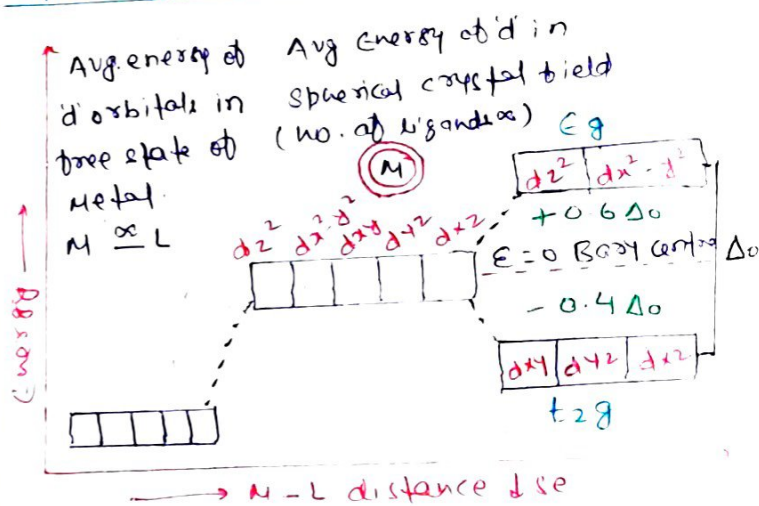


Axial

Non axial

Energy order: $-d_{x^2-y^2} > d_{xy} > d_{z^2} > d_{yz} = d_{xz}$

Splitting in octahedral



$\Rightarrow$ (Octahedral system) $\Delta_o \rightarrow$ Splitting Energy

$P \rightarrow$ Pairing Energy

$\Delta_o > P \rightarrow S.L$

$\Delta_o < P \rightarrow W.L$

$d^1 \xrightarrow{S.L/W.L} t_{2g}^1, e_g^0$

$d^2 \xrightarrow{S.L/W.L} t_{2g}^2, e_g^0$

$d^3 \xrightarrow{S.L/W.L} t_{2g}^3, e_g^0$

$d^4 \xrightarrow{W.L} t_{2g}^3, e_g^1$

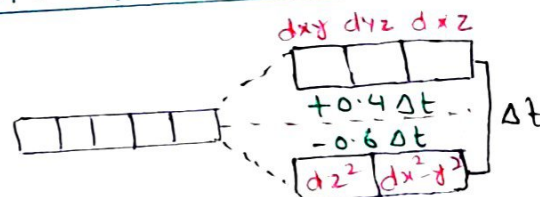
$d^4 \xrightarrow{S.L} t_{2g}^4, e_g^0$

Jugad

$W.L \rightarrow t_{2g} \rightarrow e_g \rightarrow t_{2g} \rightarrow e_g$

$S.L \rightarrow t_{2g} \rightarrow t_{2g} \rightarrow e_g \rightarrow e_g$

Splitting in tetrahedral (Octahedral)



$$\Delta_t = \frac{4}{9} \Delta_o$$

$$\Delta_t < \Delta_o$$

e-billing $\rightarrow$ Box both W.L & S.L

$e \rightarrow t_2 \rightarrow e \rightarrow t_2$ Always

$$\Delta_t < P$$

C.N=4 $\rightarrow$ Use VBT

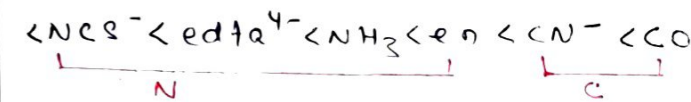
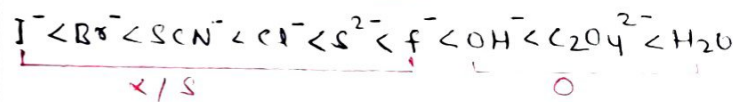
C.N=6 $\rightarrow$ VBT/CFT

Factors affecting splitting (Δ)

① Nature of Ligand

$\rightarrow$ Mainly based on donor strength
($C > N > O > X$)

Spectrochemical series

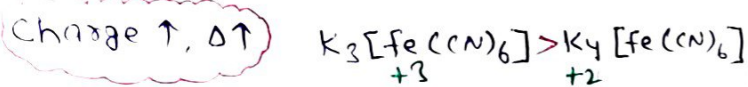


$NCS \rightarrow S.L \rightarrow N$ donor

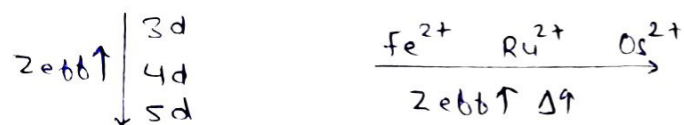
$SCN \rightarrow W.L \rightarrow S$ donor

② Charge on CNA/CNI

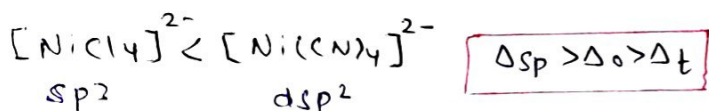
(Ligand same, Metal same)



③ Zeff of CNA/CNI (Zeff ↑, $\Delta \uparrow$)



④ Geometry $\rightarrow$ (C.N = 4, C.N = 6)



$\Delta_o = \frac{9}{4} \Delta_t$ or $\Delta_t = \frac{4}{9} \Delta_o$ $\Delta_{sp} = 1.33 \Delta_o$

$\rightarrow$ Ligand & metal must be same.

Calculation of CFSE of octahedral

CFSE $\rightarrow$ Crystal field stabilization energy

CFSE = $n(-0.4 \Delta_o) + m(+0.6 \Delta_o) + xP$

n = no. of e^- in t_{2g}

m = no. of e^- in e_g

Δ_o = splitting energy

P = Pairing energy

x = no. of new pairs

pairs formed due to presence of S.L

In case of w.l pairing energy will be zero ($\therefore x=0$)

* d^4 in w.l $t_{2g}^3 e_g^1$

$n=3$

$m=1$ $= 3(-0.4 \Delta_o) + 1(+0.6 \Delta_o) + 0$
 $= -0.6 \Delta_o$

* d^4 in S.L $t_{2g}^4 e_g^0$

$n=4$ $= 4(-0.4 \Delta_o) + 0(+0.6 \Delta_o) + 1P$
 $m=0$ $= -1.6 \Delta_o + 1P$

Jugaad

New = no. of pairs in presence of S.L
 pair = no. of pairs in w.l

* Stability of complexes - depends on

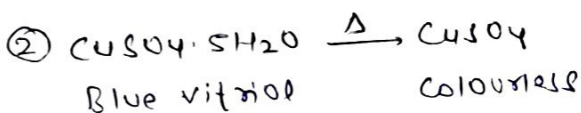
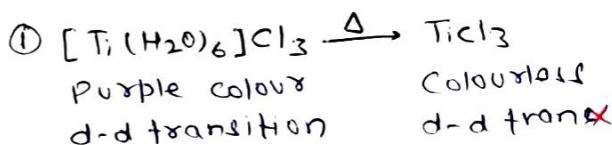
- ① charge on M
- ② strength of ligand
- ③ chelation (Ring formation)

Colour in complexes

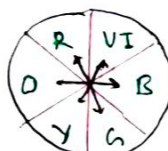
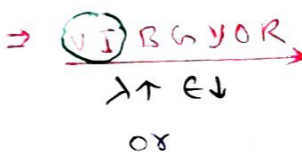
$\rightarrow$ Gen. complexes are coloured due to d-d transition $E(d_o) \propto \text{strength of ligand}$
 $\rightarrow$ Explained by CFT.

$\epsilon \propto \frac{1}{\lambda}$ λ = absorbed wavelength
 ϵ = splitting energy (Δ_o)

$\rightarrow$ Gen, d^0 & d^{10} are colourless
 $\downarrow$ d-d transition $\times$



$\downarrow$
 - out of 5 H_2O 4 are coordinated, 1 is H-bonded.



Jugaad
 $\rightarrow$ Strong ligand
 High energy vala
 Colour absorb krta hai!!

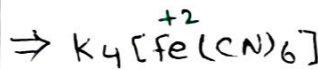
EAN (Effective Atomic number)

$EAN = Z - 0.5 + e^- \text{ gained by ligand}$

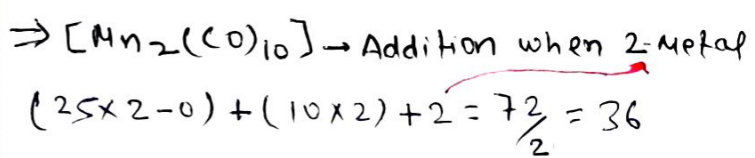
e^- ka hisaab $\rightarrow$ kitne gaaye (0.5) & kitne aaye (e^- from ligand).

EAN Rule is followed by

$\rightarrow$ Equal to next inert gas,



$EAN = 26 - 2 + (6 \times 2) = 36$



Organometallic Compounds

→ When carbon directly attached with Metal



① σ -bonded

- $RMgX$ (Grignard reagent)
- R_2Zn (Frankland)
- R_2CuLi (Gilman)
- $(C_2H_5)_4Pb$ (tetraethyl lead)
- $TiCl_4 + R_3Al$ (Ziegler-Natta catalyst)
or
 $TiCl_3 + R_2AlCl$

② π -bonded

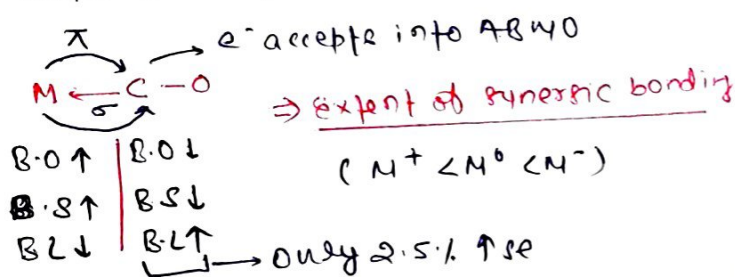
- Ferrocene
- Cobaltocene
- Ruthenocene
- Zeise's salt

③ $\sigma + \pi$ Bonded - (Metal Carbonyls)

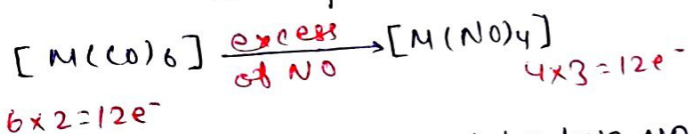
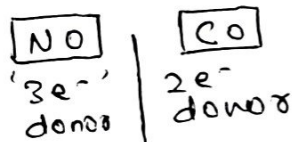
→ $Ni(CO)_4$, $Fe(CO)_5$ etc.

Bonding in Metal Carbonyls: $M(CO)_n$

Metal carbonyls → $\sigma + \pi$ bonded pairs



(NOTE)



→ 3 'Co' can be replaced by two NO.

Isomerism in Coordination compound

→ Molecular formula → Same

① Structural isomerism

- ① Ionization ② Linkage isomerism
- ③ Hydrate isomerism ④ Coordination isomerism

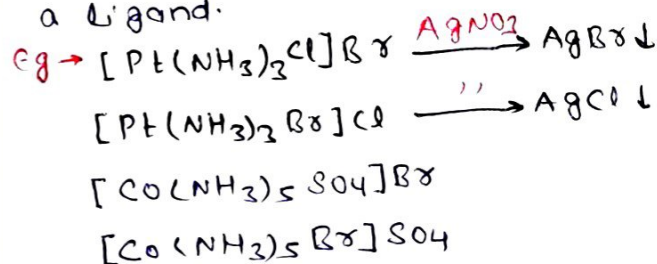
② Stereo isomerism

- ① Geometrical
- ② Optical

Structural Isomerism

① Ionization Isomerism

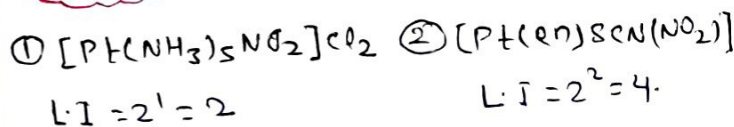
→ Produce different ions on ionization.
 → the counter ion in a complex salt is itself a potential ligand & can displace a ligand.



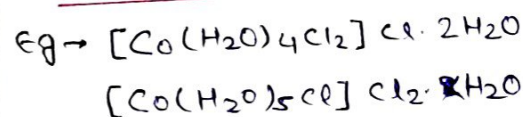
② Linkage Isomerism

→ Arises due to Ambidentate ligands.

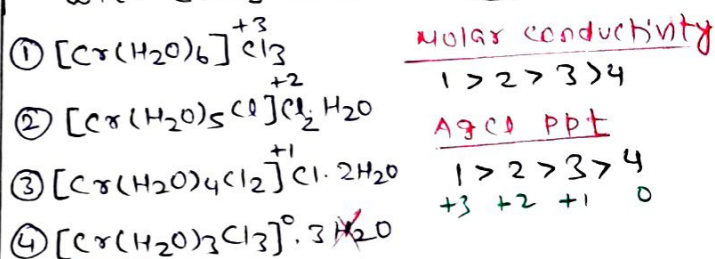
$L.I = 2^n$ n = no. of ambidentate ligands



③ Hydrate Isomerism - (Solvate isomerism)



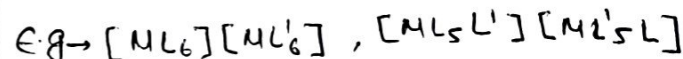
→ How many hydrated isomers are possible with $CrCl_3 \cdot 6H_2O$? → ③



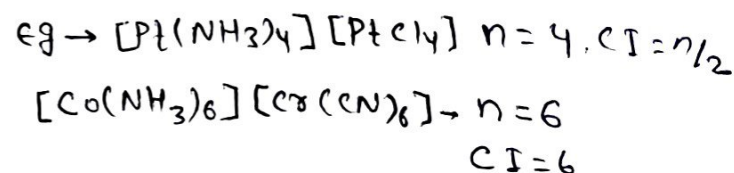
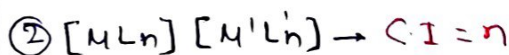
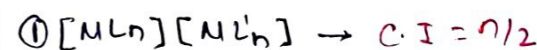
→ Reason → Complex neutral ho gya, pani bahar nahi sahega.

④ Coordination Isomerism

→ Arises due to exchange of ligands b/w Cationic and Anionic complexes.



(Jugaad) → for C.I when Monodentate ligands are present.



Isomerism → Same chemical formula & same chemical bonds but different spatial arrangements.

Geometrical Isomerism

- Arises in heteroleptic complexes.
- Cis - two ligands adjacent to each other (90°)
- Trans - " " opposite " " (180°)

Optical Isomerism :- Optical isomers are mirror images that cannot be superimposed one on another. These are called as enantiomers (optically active pairs).

Square planar complex (CN=4)

	Comp	G.I	
Optical X	MA_4	0	- all b/c d/l f
POS ✓	MA_3B	0	Monodentate ligands
G.I ✓	MA_2B_2	2	(AA) → symmetrical
	MA_2BC	2	bidentate (donor at same [en, ox])
	MA_2BED	3	(AB) → unsymmetrical bidentate (d/bb)

By using Bidentate ligands

$M(AA)_2$	Jugaad	G.I = 0	Symmetrical bidentate
$M(AA)_2B_2$			
$M(AA)_2BC$			
			(AA) ↓ G.I = 0 (always)
			(AB) for unsymmetrical bidentate
			$M(AB)_2 = 2 \cdot G.I$
			$M(AB)_2B_2 = 0$
			$M(AB)_2BC = 2 \cdot G.I$

Tetrahedral complex (CN=4)

G.I X → optically active i.d. - (i) MA_2BC (ii) $M(AB)_2$

Jugaad → (CN=4)

- ① Zn, Cd, Hg → always tetrahedral
- ② Pt, Pd → always sq. planar

Octahedral complex (CN=6)

→ can give both G.I & O.I

Jugaad → for optically active / inactive

- ① If two same ligands present at any one trans position → POS ✓, O.I
- ② If any two trans position are mutually same → POS ✓, O.I
- ③ If non co-planar rings are present → O.A

Type	No. of G.I	S.I
MA_6	0	0
MA_5b	0	0
MA_4b_2	2	2
MA_4b_3c	2	2
MA_3b_3	2	2

* Using Bidentate ligands

① $M(AA)_3 \rightarrow [Co(en)_3]^{3+}$
G.I (X) O.I (✓) S.I = 2

② $M(AA)_2b_2$

G.I = 2 cis & trans
S.I = 3 "cis is active"

③ $M(AA)_2b_2c$

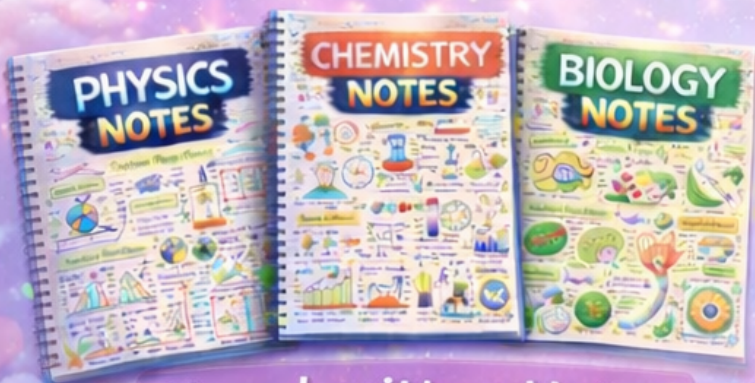
G.I = 2 cis & trans
S.I = 3 "cis is active"

Importance & Applications of Coordination compounds

- ① Hardness of water is estimated by simple titration with Na_2EDTA .
- ② In extraction & purification of metals.
- ③ Rhodium complex, $[RhCl(PPh_3)_3]$, a Wilkinson catalyst, is used for the hydrogenation of Alkenes.
- ④ In black & white photography, the developed film is fixed by washing with hypo solution which dissolves the undecomposed $AgBr$ to form a complex ion, $[Ag(S_2O_3)_2]^-$.
- ⑤ EDTA is used in the treatment of lead poisoning.
- ⑥ Cis-platin is used as anticancer.
- ⑦ Chlorophyll (pigment responsible for photosynthesis) → (Mg)
- ⑧ Haemoglobin → Fe
- ⑨ Cyanocobalamin vitamin $B_{12} \rightarrow Co$
- ⑩ Articles can be electroplated with silver & gold much more smoothly & evenly from solutions of the complexes, $[Ag(CN)_2]^-$ & $[Au(CN)_2]^-$ than from a soln of simple metal ions.

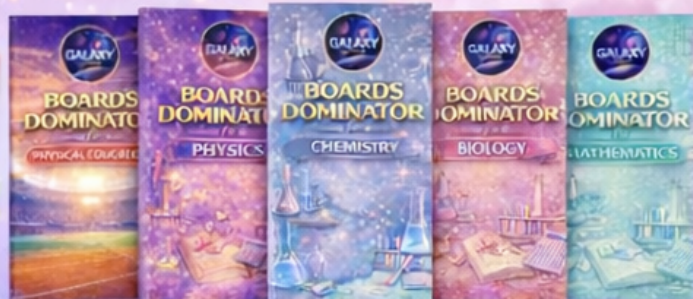
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